

Revision Execution Plan

*Significance Bias in Government Intervention Research: Evidence from
Top Finance Journals*

Response to Journal of Finance Referee Report

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How to read this document. Each referee concern is assigned a priority label: **[HIGH]** = must fix before resubmission; **[MEDIUM]** = should fix, materially strengthens the paper; **[LOW]** = desirable but optional. Each item states the *referee concern*, the *planned response*, the *specific code and paper changes* required, and the estimated *difficulty*. Items are ordered as they appear in the referee report.

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1 Major Concerns

1.1 Concern 1 — Fundamental Identification Problem: Low NBER-to-Journal Match Rate **[HIGH]**

Referee concern. Only 8 of 183 published papers (4.4%) are matched to an NBER working paper. The sample therefore compares two largely non-overlapping populations. Null-finding papers may have lower publication rates because they are of lower quality, not because of a significance filter. The referee requests either (a) a higher match rate, (b) controls for observable quality, or (c) a substantially qualified interpretation.

Planned response — three-part strategy. Part A: Improve the NBER-to-journal match rate. The current fuzzy-title matching (RapidFuzz token-sort, threshold ≥ 90) misses pairs where titles were significantly revised during editorial review. We will implement three additional matching passes:

1. *DOI-based matching via CrossRef.* For each in-scope journal article, query the CrossRef API for the paper’s DOI and retrieve the `relation.is-preprint-of` field, which CrossRef populates when a journal paper explicitly links to a working paper version. This covers cases where the journal paper itself declares the NBER link.
2. *OpenAlex “related works” matching.* Each OpenAlex record has a `related_works` field containing linked pre-prints and working paper versions. We will query the OpenAlex API for each in-scope journal paper and check whether any related work is an NBER working paper (identifiable by the NBER URL pattern).
3. *Author-year-keyword matching.* For journal papers with no title match, extract the first author’s surname and the publication year; then search the NBER working paper database for papers by the same first author in the same year that share at least two keywords with the journal paper. Accept matches with a keyword overlap score ≥ 2 .

Code change: Add a new script `05b.enhanced_matching.py` implementing these

three passes. Expected improvement: match rate from 4.4% to approximately 15–25% of published papers.

Part B: Add quality controls to the probit. Even after improving matching, most published papers will lack NBER antecedents. We therefore add the following quality controls to the probit specification, all of which are computable from existing data:

- *NBER citation count* (already in the dataset: `nber_citations`). Higher-cited working papers are of higher quality and more likely to be published regardless of findings.
- *Top-institution indicator* (already in the dataset: `top_institution`). Papers from top-10 finance departments face a different editorial environment.
- *Number of authors* (`n_authors`). Multi-author papers tend to have more diverse networks and higher publication rates.
- *LLM-coded identification strategy*. We will add a new LLM coding pass (script `04b_code_identification.py`) that classifies each paper’s primary identification strategy as one of: difference-in-differences (DiD), regression discontinuity (RD), instrumental variables (IV), event study, panel OLS, or cross-sectional OLS. DiD and RD papers are, on average, of higher identification quality and should be controlled for.

Paper change: Expand equation (1) to include the quality controls $\mathbf{X}_i$ substantively (currently they are mentioned but unused in the baseline). Report the quality-controlled specification as the new primary specification; move the bivariate probit to a robustness check.

Part C: Strengthen the identification discussion. Add a subsection to Section 3 explicitly framing the design as a “revealed-preference” comparison: we observe the direction distribution in the NBER pool (the supply of research) and the direction distribution in the published sample (the demand-selected output), and the gap between them is the significance filter. Cite comparable designs in the meta-science literature (Franco et al., 2014; Dwan et al., 2008) to show this approach is standard. Acknowledge explicitly in the limitations section that the design is observational, not

experimental, and that unobservable quality differences could attenuate or amplify the estimated premium.

Difficulty: High. Part A requires new API queries and matching logic (~ 2 days of coding). Part B requires one new LLM coding pass and rerunning `06_analysis_main.py` (~ 1 day). Part C is a writing task (~ 0.5 days).

1.2 Concern 2 — NBER Selection Bias [MEDIUM]

Referee concern. NBER researchers are elite. Their null-finding papers may be more publishable than the average null result in the field. The 36% null publication rate may therefore overstate the publication probability for null results generally.

Planned response. Part A: Bound the bias. We add a discussion in Section 6 (Limitations) that formalizes the NBER selection concern. If NBER researchers have a 20% quality premium over the average researcher, the true null-paper publication rate outside NBER would be $\approx 0.361/1.20 = 30\%$, implying a directional premium of $82\% - 30\% = 52$ pp rather than 46 pp. In other words, NBER selection makes our estimate *conservative*. We should make this point explicitly: the bias from NBER selection, if any, works *against* our findings, not in their favor.

Part B: Placebo test using non-intervention NBER papers. We will run the full pipeline on a comparison sample of NBER working papers from the same five programs that do *not* match any government intervention keyword. If NBER selection were driving our findings, we would expect a similar direction–publication-rate relationship in this placebo sample. If instead the null-paper penalty is concentrated in the government- intervention papers, NBER selection is not the mechanism.

Code change: Modify `06_analysis_main.py` to build and report the placebo sample. *Paper change:* Add placebo results as Robustness Check R6.

Difficulty: Medium. The placebo sample requires running direction coding on a random draw of ~ 200 non-intervention NBER papers, which requires one additional LLM pass (~ 1 day).

1.3 Concern 3 — LLM Validation Statistics [HIGH]

Referee concern. No human validation statistics are provided. The referee notes three sub-concerns: (a) no confusion matrix or inter-rater agreement; (b) published abstracts are more polished than NBER abstracts, so the LLM may systematically assign stronger direction codes to published papers; (c) the definition of “positive” is normatively charged.

Planned response. Part A: Manual validation of a random sample. We will manually code the direction of a random sample of 60 papers (20 positive, 20 negative, 20 null as assigned by the LLM), drawn stratified by direction code and published/NBER status. Each paper will be coded by the lead author and one research assistant independently, then reconciled. We will report:

- Confusion matrix (LLM code $\times$ human code), 3×3 .
- Cohen’s κ for direction coding agreement (LLM vs. human).
- Precision and recall for each direction class.

A target of $\kappa \geq 0.70$ (“substantial agreement”) is considered acceptable for this type of classification task. If $\kappa < 0.70$, we will re-examine the direction coding prompt and consider a revised LLM pass.

Paper change: Add a validation paragraph to Section 3.3 (Direction Coding) and include the confusion matrix as Table A3 in the Appendix.

Part B: Test for differential abstract polish. The referee’s concern that published abstracts are more decisive in their framing is a legitimate measurement threat. We will test this directly by comparing the LLM’s *confidence* scores (high/medium/low) across published vs. NBER papers. If published paper abstracts systematically elicit higher LLM confidence, this supports the abstract-polish concern. We will also run the direction coding on the *NBER* version of matched papers (for the 8 matched pairs) and compare direction codes to confirm consistency.

Part C: Alternative positive/negative convention. We add a robustness check (R7) that collapses positive and negative into a single “directional” indicator ($\text{Directional}_i = \mathbf{1}[d_i \neq 0]$) and estimates a bivariate probit of *Published* on *Directional*.

This sidesteps the normative-framing concern entirely: the only distinction is directional vs. null, and the sign of the direction is irrelevant. The expected coefficient on *Directional* is the average of the current $\hat{\beta}_1$ and $\hat{\beta}_2$, approximately 1.30***.

Code change: Add the directional indicator and re-run `06_analysis_main.py`.

Paper change: Add R7 to the robustness table and a paragraph in Section 5 (Robustness).

Difficulty: Medium. The manual validation takes approximately 1 day of human coding time. The additional LLM confidence analysis and R7 are straightforward (~ 0.5 days of coding).

1.4 Concern 4 — Quality Controls [HIGH]

Referee concern. Null-finding papers may have lower publication rates because they are of genuinely lower empirical quality, not because of a significance filter. The paper provides no controls for identification strategy, author quality, data quality, or statistical power.

Planned response. This concern overlaps with Concern 1 Part B above. The specific controls we will add are:

1. **Identification strategy** (LLM-coded, script 04b): $\text{DiD} = 1$, $\text{RD} = 1$, $\text{IV} = 1$, $\text{event study} = 1$, $\text{panel OLS} = 0$. These are coded from the abstract and represent the most observable signal of identification rigor.
2. **NBER working paper citations** (already available: `nber_citations`): log-transformed, captures paper quality as recognized by other researchers before publication.
3. **Top-institution indicator** (already available: `top_institution`): 1 if any author is at a top-10 finance department.
4. **Number of authors** (already available: `n_authors`): more authors may signal more resources and higher average quality.

We will show that the significance-bias coefficients ($\hat{\beta}_1, \hat{\beta}_2$) are stable in magnitude and significance after adding all quality controls. If the coefficients *shrink* substantially after quality controls, we will qualify the interpretation accordingly.

Code change: Modify `06_analysis_main.py` to include the new controls in the extended specification. *Paper change:* Add a new column to Table 3 (“+ Quality Controls”) and discuss the results in Section 4.2.

Difficulty: Low-Medium. The identification strategy coding requires a new LLM pass (~ 0.5 days); adding controls to the probit is straightforward (~ 0.5 days).

1.5 Concern 5 — JF Coefficient Labeling Inconsistency **[HIGH]**

Referee concern. Table 4 labels the JF coefficients as 1.022*** and 1.067*** (indicating $p < 0.01$), but the text simultaneously states they “fall short of statistical significance at conventional levels.” This is a direct contradiction.

Planned response. This requires a factual verification. The `table4_probit_by_journal.csv` output shows:

- JF Positive: coefficient = 1.0222, p-value = 0.0006 $\Rightarrow$ **significant at $p < 0.001$.**
- JF Negative: coefficient = 1.0673, p-value = 0.0004 $\Rightarrow$ **significant at $p < 0.001$.**

The triple-star notation in Table 4 is **correct**. The text is **wrong**. The sentence “fall short of statistical significance at conventional levels” must be deleted and replaced with an accurate characterization: the JF coefficients are statistically significant but *smaller in magnitude* than those for RFS and JFE. The discussion should focus on the economic magnitude difference (1.022 vs. 1.229 and 1.420) rather than falsely claiming non-significance.

Paper change only. Edit Section 4.3 (Cross-Journal Analysis) and the corresponding paragraph in the Introduction.

Difficulty: Very Low. Approximately 30 minutes of text editing.

2 Moderate Concerns

2.1 Concern 6 — Small Samples and Linear Probability Models [MEDIUM]

Referee concern. Sample sizes in sub-group analyses are small (JF: 38 published papers; sub- periods: 97 and 154 papers). The referee requests linear probability model (LPM) results as a robustness check against probit separation issues.

Planned response. Add LPM estimates to the robustness table (as R8) for the main specification and the by-journal specification. LPM coefficients are directly interpretable as marginal effects and do not suffer from perfect separation. We also add heteroskedasticity-robust standard errors (the probit already uses sandwich SEs; confirm this). Bootstrap confidence intervals (1,000 replications) will be added for the sub-period estimates where sample sizes are smallest.

Code change: Add LPM estimation to `06_analysis_main.py` using `statsmodels.OLS`.

Paper change: Add R8 row to Table 5 (Robustness).

Difficulty: *Low.* Approximately 1 hour of coding.

2.2 Concern 7 — Report the 2000–2007 Sub-Period [MEDIUM]

Referee concern. The 2000–2007 sub-period (45 papers) is omitted because of small sample size. The referee argues 45 papers is adequate for a bivariate probit and requests results.

Planned response. Run the bivariate probit on the 2000–2007 sub-period and add results to Appendix Table A2. Given the small N , report (a) probit estimates with a prominent small-sample caveat, (b) Fisher’s exact test as an alternative non-parametric test, and (c) raw publication rates (positive, negative, null) for transparency. We do *not* move these results to the main text but report them fully in the appendix.

Code change: Modify `06_analysis_main.py` to include the 2000–2007 period in the sub-period loop, removing the 45-observation exclusion threshold. *Paper change:* Update Appendix A.3 and add a footnote in Section 4.4 pointing to the appendix.

Difficulty: Very Low. The code change is a one-line threshold adjustment (~15 minutes). The paper change is ~30 minutes.

2.3 Concern 8 — Corpus Count Inconsistency [DONE]

Referee concern. The Introduction mentions “more than 40,000 documents” but the Data section reports 33,224.

Planned response. This discrepancy already exists in the current draft. The Introduction references an older draft figure (pre-2000 restriction). **Fix immediately:** search for “40,000” in `main.tex` and replace with “33,000” or the precise figure “33,224.”

Paper change only. Edit one sentence in Section 1 (Introduction) and one sentence in Section 3.3.

Difficulty: Very Low. Approximately 10 minutes.

2.4 Concern 9 — Replace Robustness Check R4 [MEDIUM]

Referee concern. R4 restricts to high-confidence LLM classifications but is endogenously selected (88% publication rate vs. 62% in the full sample). The authors acknowledge it is uninformative, so it should be replaced.

Planned response. Drop R4 and replace it with two new, substantive robustness checks:

- **New R4: Directional vs. Null Indicator (binary specification).** Collapse the direction code into a single binary $\text{Directional}_i \in \{0, 1\}$ and estimate a bivariate probit. This addresses Concern 3(c) about the positive/negative coding convention and also provides a simpler, cleaner test of the core hypothesis.
- **New R5: Exclude 8 matched papers.** Drop the 8 NBER-to-journal matched pairs to ensure results are not sensitive to these ambiguous observations (this check already exists in the text as R5; formalize it in the table).

Code change: Update `06_analysis_main.py` and `08_robustness.py` to add the binary-directional specification and remove the high-confidence filter check. *Paper change:* Update Table 5 (Robustness) and Section 5.

Difficulty: *Low.* Approximately 2 hours of coding and writing.

3 Minor Comments

3.1 Minor 1 — Quantify Unmatched Published Papers [MEDIUM]

Planned response. Run a manual spot-check of 20 randomly selected in-scope journal papers that have no NBER match. Assess (by author affiliation and Google Scholar search) whether they likely had NBER working paper antecedents. Report the spot-check findings in a footnote to provide the reader with a sense of the true match rate under a more complete search.

Difficulty: *Low-Medium.* Approximately 2 hours of manual checking.

3.2 Minor 2 — Comparison Domain (Placebo) [MEDIUM]

Planned response. As described in Concern 2 Part B, we will construct a placebo sample of non-intervention NBER papers from the same five programs. If the direction-publication premium is absent in this placebo group, it confirms the finding is specific to government-intervention research, not an artifact of NBER selection. This check doubles as a response to both Minor 2 and Concern 2.

3.3 Minor 3 — Formalize the Mechanism Discussion [LOW]

Planned response. Add a structured “mechanism tests” subsection to Section 6. While we cannot access editor letters or rejection data, we can provide indirect evidence:

- If the bias is purely editorial (demand-side), it should be uniform across authors with different network strengths. We can test this by interacting the direction

indicators with the top-institution flag: if the premium is larger for non-top-institution authors, this is consistent with a quality-confound story; if it is uniform, the significance-filter story is supported.

- The time variation (negative premium during 2008–2015) is more consistent with an editorial-demand mechanism than a researcher- strategic mechanism. Formalize this argument with a brief test: if the mechanism is researcher-strategic, we would expect the *submission* rate of null papers to fall during 2008–2015. Without submission data we cannot test this directly, but we can check whether the *share* of null-finding NBER papers changed in this period.

Code change: Add interaction term to `06_analysis_main.py`. *Paper change:* Add a mechanism paragraph in Section 6.2.

Difficulty: *Low*. Approximately 1 hour of coding and 1 hour of writing.

3.4 Minor 4 — Multiple Testing Correction [LOW]

Planned response. Add a paragraph in Section 5 (Robustness) noting that our main hypothesis is pre-specified (the direction $\times$ publication relationship), and that the sub-group and sub-period tests are exploratory. Report Bonferroni-corrected p -values for the six robustness specifications in Table 5. Because our main coefficients have $p < 0.001$ and the robustness checks cover six tests, the Bonferroni-corrected threshold is $0.05/6 \approx 0.008$, which all main results easily clear.

Paper change only. Approximately 30 minutes.

3.5 Minor 5 — Terminology [LOW]

Planned response. Retain the term “significance bias” but add a definitional sentence in the Introduction clarifying its meaning in the context of this paper: “*By significance bias we mean the tendency of journals to publish papers with any directional (statistically significant) result at higher rates than papers with null or inconclusive results, regardless of the sign of the estimated effect.*” This distinguishes our usage from the test-statistic heaping literature.

Paper change only. One sentence.

3.6 Minor 6 — Update LLM References [\[LOW\]](#)

Planned response. Add two or three recent references on LLM reliability in economics classification tasks to the Related Literature section. Candidates include: Korinek (2023) on LLMs in economics research; Aguirre et al. (2023) on GPT-4 for text annotation; and Lopez-Lira and Tang (2023) on LLMs for sentiment analysis in finance. Update `references.bib` accordingly.

Paper change only. Approximately 30 minutes.

4 Summary and Prioritized Work Plan

Table 1 organizes all planned changes by priority and estimated effort.

Table 1: Prioritized Revision Work Plan

Item	Concern	Action	Type
1	Fix JF significance text (C5)	Correct text in Sections 1 and 4.3	Paper
2	Fix “40,000” corpus text (C8)	Replace with 33,224 in Introduction	Paper
3	Add 2000–2007 sub-period (C7)	Remove threshold in script 06	Code + Paper
4	Add LPM robustness (C6)	Add OLS to script 06	Code + Paper
5	Replace R4 with binary spec (C9)	New specification in script 06	Code + Paper
6	Strengthen identification text (C1C)	Rewrite Section 3 framing	Paper
7	Add quality controls (C4, C1B)	Add controls to probit script 06	Code + Paper
8	LLM identification coding (C4)	New script 04b + re-run 06	Code + Paper
9	Manual direction validation (C3A)	Human-code 60 papers + kappa	Human + Paper
10	Test abstract polish (C3B)	Compare LLM confidence by source	Code + Paper
11	Add binary direction robustness (C3C)	Directional vs. null indicator	Code + Paper
12	Enhanced NBER matching (C1A)	New script 05b with 3 API passes	Code + Paper
13	NBER selection bound (C2A)	Add bounding argument to Sec. 6	Paper
14	Placebo test (C2B, Minor 2)	Direction coding on non-intervention WPs	Code + Paper
15	Mechanism interaction test (Minor 3)	Add top-institution interaction	Code + Paper
16	Manual spot-check unmatched (Minor 1)	Check 20 papers by hand	Human
17	Multiple testing paragraph (Minor 4)	Add Bonferroni discussion	Paper
18	Terminology clarification (Minor 5)	Add definitional sentence	Paper
19	Update LLM references (Minor 6)	Add 2–3 citations to .bib	Paper
Total estimated effort			

Recommended Sequence

Week 1 — Quick wins and data corrections: Items 1–7 above. These are all either text corrections or short code additions using data already in hand. They address two of the six “required revision” items (JF text fix, 2000–2007 sub-period, LPM check) and improve the robustness section significantly.

Week 2 — New coding passes: Items 8–11. Run the identification-strategy LLM coding pass (script 04b), add quality controls to the probit, conduct the manual direction validation, and add the abstract-polish test and the binary-direction robustness check.

Week 3 — Enhanced matching and placebo test: Items 12–14. Implement the three-pass NBER-to-journal matching enhancement (CrossRef, OpenAlex, author–year–keyword), run the placebo sample for the non-intervention NBER papers, and formalize the NBER selection bounding argument.

Week 4 — Polish and paper writing: Items 15–19 plus full paper rewrite incorporating all new results. Recompile, proofread, and prepare the cover letter summarizing responses to each referee concern.

What We Are *Not* Pursuing

The referee suggests obtaining “editor records, submission-level data from a journal, or desk-rejection data.” These data are not publicly available and would require multi-year cooperation with journal editors. We will explicitly note this limitation in the cover letter and explain that our approach follows the established design in the meta-science literature (Franco et al., 2014; Dwan et al., 2008), which similarly relies on a pre-registered or pre-observable research pool rather than submission records.

This document will be updated as revisions are completed. Completed items should be moved to the DONE list in the cover letter.